



Indian Institute of Information Technology, Una [HP]

An Institute of National Importance under MoE

Saloh, Una (HP) - 177209

Website: www.iiitu.ac.in

Date: 17/01/26

PROJECT PLAN PROJECT PHASE - II

Student Name	Swayam Shah Sampurn Samadder Gyanendra	Roll No.	23254 23247 23222
Branch	ECE	Supervisor	Dr. Naveen Cheggoju

1. Title of the Project

WorkPulse - AI-Powered Multi-Tenant Team Management SaaS Platform

2. Introduction

This project involves the design and development of a web-based team management platform powered by Artificial Intelligence. The application is built using the MERN stack and follows a multi-tenant architecture, allowing multiple organizations to use the same application while keeping their data completely isolated.

The system enables companies to manage teams, assign tasks, monitor progress, and generate insights. It also includes AI-based features such as PDF summarization, automated analysis of Excel files, and graphical representation of data using charts and pie diagrams. The AI assistant helps organizations understand productivity trends and improve workflow efficiency.

3. Problem Definition

Many organizations rely on multiple tools for team management, document analysis, and reporting. This leads to fragmented workflows, lack of centralized insights, and reduced productivity. Traditional project management tools often lack AI-driven analytics and automated reporting features.

There is a need for a unified, intelligent platform that:

- Manages teams and tasks in one place
- Provides automated insights using AI
- Summarizes documents and reports
- Generates visual analytics from data
- Supports multiple organizations securely in a single system

This project addresses these needs by building a scalable, AI-integrated, multi-tenant SaaS solution.

4. Objectives

- To design and develop a multi-tenant SaaS platform using the MERN stack.
- To implement secure user authentication and company-based data isolation.
- To build team and task management features.
- To integrate AI-based PDF summarization.
- To analyze Excel data and generate charts and pie graphs.
- To create an AI assistant that provides productivity insights.
- To deploy the platform on a cloud environment.

5. Skillset additionally required to solve/address the problem

- JavaScript (ES6+) programming
- React.js for frontend development
- Node.js and Express.js for backend APIs
- MongoDB and Mongoose for database management
- JWT authentication and authorization
- REST API design and integration
- Multi-tenant system architecture
- File handling (PDF and Excel parsing)
- Chart libraries (Chart.js or Recharts)
- AI API integration
- Cloud deployment (Vercel, Render, or similar)

